

Hospital

Case study:

A hospital has hired you to **install a new wireless network**. The hospital currently has a wireless network that is a few years old. **The problem with the current wireless network is that it was built using proprietary standards instead of 802.11.**

This current wireless **network does not have an upgrade path to 802.11**

On the new wireless network, the hospital wants a system that **allows nurses to scan a bar code on the patient's wristband and then have all the patient's information displayed on a handheld device.**

The information displayed should include:

- 1.Patient's name
- 2.Age
- 3.Medication
- 4.Diet
- 5.Any past or future surgeries
- 6.Name of the patient's doctor

They would also like to have **wireless IP phones that doctors can carry while on duty** in case of emergencies. The Chief Information Officer (CIO) has asked you to come up with a plan to implement this new wireless network.

Directions: Looking at the current wireless network and what the hospital would like to accomplish with the new network, **explain to the hospital administration why the current wireless network will have to be replaced before you can begin the project.**

802.11 Case Study- Hospital

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11/11/2023

The Problem with the Current Wireless Network

To digitally upgrade the hospital for efficiency and safety, it is imperative that the hospital replaces the current WLAN with the latest 802.11 standard. As financially beneficial as it would be to stay with the current wireless network and upgrade it to meet the goals that the hospital has in mind, it is simply not possible. The hospital's goals are to implement wireless devices that will immediately scan and access the most recent patient records and wireless IP phones for doctors to carry on duty in case of emergencies. These two requests required network services that are fast and have immediate/priority delivery, have encryption and other security capabilities and that allow for a wide range of mobility of its users within a certain location range. The current state of the hospital's wireless network is built using the proprietary standards which make it impossible to update and discourages the inclusion of additional vendors for new devices. Due to the current network's inability to upgrade to the 802.11 standard it is imperative that the network is completely switched over to the 802.11 standard in order to increase speed and provide all of the necessary resources the hospital has the goals to provide.

The Benefits of global/industry standards vs. proprietary standards

Proprietary standards offered a unique and industry-focused perspective when developing a wireless network standard for healthcare and hospitals as a whole. It allowed for the basic networking services to be performed as needed, but with the current digital needs that hospital has, the upgrades, additions, changes and security needs that are to occur can only be done through the global standardization 802.11. For instance, 802.11 standards as prepared by the IEEE, provides wireless connectivity for fixed, portable, and moving stations within a local area. In addition, this standard supports asynchronous and time-bounded delivery, continuity of service within extended areas, multiple high speed transmission, network management and

authentication/encryption services. It also allows for power management, bandwidth efficiency and other security features.

Avoid Mixed Standards

Though it is tempting, the institution is to avoid mixing standards in every means possible. Mixing standards is simply a band aid solution when looking at the various devices, network needs and services that the hospital needs in order to provide safe, secure, efficient, and accurate care to their patients and medical staff. If the institution was to try and mix a standard with the current proprietary standard, the networks will be more expensive to maintain and completely inflexible. Along with the various components it needs to manage and segment, mixing standards will require multiple separate networks to run from separate vendors without the ability to share appliances and will require all relevant personnel to be proficient in the different operations between the multiple types of networks being used. This does not create a safe, efficient and logical space for hospital patients and the staff.

Plan to Correct the Problems with the Current WLAN

The current wireless network is deemed to be incompatible with the goals that the hospital has for its digital future. The first step is to gather as much information about the current network, the hospital, the devices used in the hospital, the physical layout of the hospital, the policies and regulations the hospital has that may impact the WLAN, etc. The second step is to interview the users of the network to see where the biggest needs, problems, discrepancies, are in order to make their job more efficient and easier. Where do they need reliable signal strength or coverage? How many users will need support in what space? The third step is to analyze the requirements for the WLAN such as applications, devices, utilization, mobility, aesthetics, etc.

The fourth step is to get feedback on the requirements list and seek approval from the hospital stakeholders and relevant staff. The fifth is to take all of this information into account and start developing 3 different programs. One focused solely on smoothly transitioning the current network to the 802.11 standard WLAN, the second solely focused on integrating the application and handheld network device, and the third focused on integrating the IP phones.

References

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